



UNIVERSITY *of*
DEBRECEN

3D game development with Unity

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Problem posing:

- Balancing storytelling and interactive gameplay.
- Highlighting real-world career struggles.

Tasks:

- Develop a 3D first-person game with cooking mechanics.
- Integrate a 2D visual novel introduction.



Methods and Technologies

Tools Used:

- Unity (Game Engine)
- Krita (2D Art)
- C# (Scripting)

Methods:

- **State Machines:** For cooking stages (raw → cooking → burned).
- **Raycasting:** Detect and interact with kitchen objects (e.g., stove, counters).
- **Randomization:** Order system generates unpredictable customer requests.



Visual Novel Introduction

2D Storytelling:

- Mei's journey as a struggling computer science graduate.
- Visual novel style implemented using custom dialogue systems.

Assets:

- Character sprites and backgrounds designed in Krita.

Integration:

- Transition from visual novel to gameplay for a seamless player experience.



Cooking Simulation Mechanics

Core Gameplay:

- Prepare food orders (e.g., burgers, salads) based on dynamic requests.
- Manage stations: grill, fryer, and assembly counter.

Cooking Workflow:

- Food stages: Raw → Cooking → Cooked → Burned.

Challenges:

- Timing and precision in completing orders.



First-Person Controls

Character Movement:

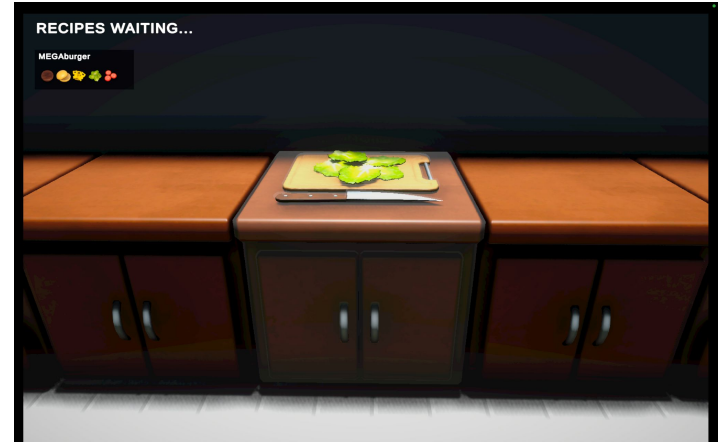
- WASD for movement, mouse for looking around.
- Smooth navigation implemented with Unity's **Character Controller**.

Object Interaction:

- Raycasting to detect and interact with kitchen items (e.g., stoves, ingredients).

Feedback System:

- Visual cues (highlighting objects) and audio feedback when interacting.



Clown Encounter

Trigger Event:

- After completing two orders, the clown appears unexpectedly.

Implementation:

- Scripted **enemy spawn** and movement logic.
- Raycasting for detection and interaction.

Purpose:

- Break the monotony and add an emotional shock to the gameplay.

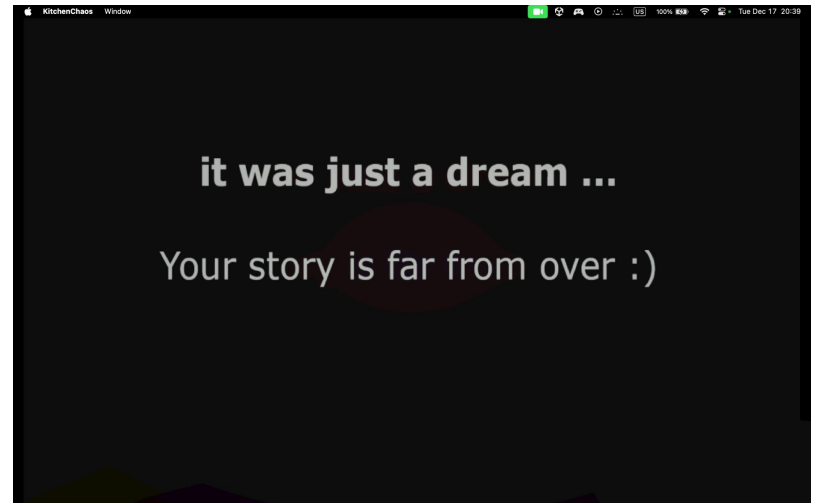
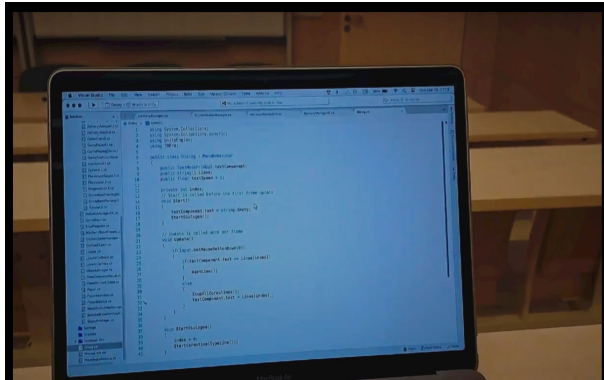


Game Ending and Motivational Message

Ending:

- Mei wakes up: It was a dream.
- Motivational message: *“Your journey is far from over.”*

Goal: Inspire players to persevere through challenges.



Conclusions and Future Work

Conclusions:

- Successfully combined storytelling and interactive first-person mechanics.
- Developed immersive cooking simulation and dynamic gameplay.

Future Work:

- **Multiplayer Mode:** Cooperative gameplay in the kitchen.
- **VR Support:** Full immersion for cooking tasks.
- **Expanded Storyline:** Branching dialogue options for multiple endings.

Game play



Thank you!